

Ahmed Ikram | Software Engineering

Dundee, Scotland | +44 7421 996048 | ahmedikram30@gmail.com | [LinkedIn](#) | [GitHub](#) | [Online Portfolio](#)

Final-year BSc Computer Science (Data Science & AI) student at the University of Dundee, on track for a First-Class degree, graduating June 2027. Designs and ships production systems end-to-end, picking the pattern that fits the load: a Rust extension where Python was the bottleneck, ARM64 where it cuts compute cost 20–30% at equal performance. An 11-service event-driven backend and a full-stack real-time platform, both OIDC-deployed with 1,400+ tests apiece. Seeking a graduate Software Engineering role.

EDUCATION

University of Dundee - BSc (Hons) Computer Science (Data Science and AI)

Sep 2023 - Present

- On track for a **First-Class Honours** degree. Expected Graduation: **June 2027**

TECHNICAL SKILLS

Languages: Python · TypeScript · JavaScript · SQL · Rust · Bash

Frontend: React · React Native · Tailwind CSS · Vite · Socket.IO

Backend & APIs: Flask · FastAPI · Node.js · REST API Design · Apache Kafka · Gunicorn · Apache Airflow · dbt · Pydantic

System Design: Microservices · Event-Driven Architecture · Distributed Systems · Idempotent Consumers · Caching Strategies · Rate Limiting · Load Balancing · Fault-Tolerant Design

Security: JWT · OAuth 2.0 · RBAC · bcrypt · AWS Secrets Manager · Encryption at Rest

Databases: PostgreSQL · Redis · SQLite · MySQL · MongoDB · Alembic

Cloud & DevOps: AWS (ECS Fargate, EC2, S3, RDS, ElastiCache, WAF, IAM, CloudFront, SageMaker, Bedrock, Secrets Manager) · Azure (Databricks, Azure SQL, ADLS) · Terraform · Docker · GitHub Actions · OIDC · Linux

Observability & Testing: Grafana · Prometheus · StatsD · Pytest · Vitest · Jest · Playwright · Cypress · React Testing Library · dbt Tests · Mutation Testing (mutmut) · Property-Based Testing (Hypothesis)

PROJECTS

DevSync Project Management Platform with GitHub Integration

React · Tailwind · Python · Flask · PostgreSQL · Socket.IO · GitHub OAuth · AWS · Docker · GitHub Actions CI/CD

- Full-stack collaboration platform: Flask REST API over a 12-table indexed PostgreSQL schema, and a JWT-authenticated Socket.IO layer broadcasting live task updates and dashboard refreshes with zero cross-project leakage; Gunicorn's gevent worker lets WebSocket and HTTP traffic share one async-I/O process without async-ifying the whole codebase.
- Three-tier RBAC via route decorators, server side GitHub OAuth CSRF validation and encrypted tokens; bidirectional Issue/PR ↔ task sync keeping DevSync and GitHub state consistent in both directions.
- AWS deployment on a custom three-tier VPC (ALB → ECS Fargate → private RDS) with OIDC-federated CI/CD and zero-long-lived credentials; a multi-stage Docker build cut the runtime image 45% (600MB → 330MB) holding sub-300ms p99 latency.
- Guarded by 1,452 automated tests (Pytest + Jest) at an 85%/85% coverage gate, CodeQL security scanning and dependency auditing - any PR that drops coverage or fails a check is rejected before it reaches deployment.

ATM Log Aggregation, Analysis & Diagnostics Platform

FastAPI · React · PostgreSQL · Kafka · Redis · ChromaDB · Model Context Protocol (MCP) · Docker · Terraform · AWS

- Backend engineering lead on a 7-person Agile team for NCR Atleos: independently architected a 11-service Dockerised platform, full AWS production deployment, and a 3-pipeline CI/CD system as a post-deadline extension on the team-delivered foundation.
- FastAPI REST API (30 endpoints, 6 routers, JWT + Redis-blacklist auth) with health-check-cascading startup across Kafka, Redis, PostgreSQL and a React frontend shipped as a ~25MB nginx-alpine image; Kafka Kraft consumer gives at-least-once delivery with idempotent deduplication and a Redis Stream DLQ.
- A standalone MCP tool server (FastMCP, SSE) exposes 12 tools over the network, reachable by any MCP client for direct inspection and debugging
- Deployed via Terraform: multi-AZ VPC (2 public + 2 private subnets), ECS Fargate behind an ALB, an EC2 host running Kafka/Redis/ChromaDB/Ollama in a private subnet, CloudFront serving the React SPA from S3.
- 3-pipeline CI/CD system: CI runs 1,438 tests (959 pytest, 394 vitest, 10 Playwright E2E, 75 Terraform assertions) and Checkov IAC compliance on every PR, then rolls out zero-downtime via OIDC with zero long-lived credentials

StockLens FinTech Spending & Investment App

React Native · Python · FastAPI · PostgreSQL · Redis · Rust · Terraform · Docker · AWS

- Async FastAPI backend: 59 REST API endpoints across 14 routers with biometric-backed JWT auth, bcrypt hashing, rate limiting and retry/backoff on the LLM agent endpoint.
- Native Rust extension (PyO3/Maturin, 12 functions) replaces a pandas bottleneck in technical-indicator computation with zero-cost abstractions, cross-built for ARM64 via QEMU in CI and called directly by both the LSTM training pipeline and live API
- Deployed via Terraform (14 modules, 193 resources): three-tier network isolation (WAF-protected ALB → ECS Fargate ARM64 → private RDS/Redis) chosen for a 20-30% compute-cost cut over x86 at equal performance, with CPU/RPS-triggered autoscaling and zero-downtime rolling deploys gated on ECS health checks.
- 1,415 backend and 822 frontend tests at a 90%/90% coverage gate, including property-based tests on financial invariants and nightly mutation testing on critical paths; enforced by a 9-job CI and a 7-stage OIDC-authenticated CD pipeline

EXPERIENCE

Beaully Mini Market, Dundee, Scotland

Retail Assistant

September 2023 - Present

- Troubleshoot and resolved Point-of-Sale and stock system errors quickly to minimise customer wait time

Zulus Peri Peri Grill, Dundee, Scotland

Front of House Worker

June 2021 - September 2023

- Trained 5 new staff members, helping them get up to speed quickly during busy service periods

CERTIFICATES

- | | |
|--|-----------------|
| • <u>AWS Academy Graduate - Machine Learning Foundations</u> | <i>Feb 2025</i> |
| • <u>AWS Academy Graduate - Cloud Foundations</u> | <i>Mar 2024</i> |
| • <u>Foundations of Azure AI: Concepts, Capabilities, and Implementation</u> | <i>Oct 2024</i> |